WORKED SOLUTION

STEP III 2006, Question 3

Pure • Power Series Coefficients

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q3

The key idea

We never have to differentiate $\tan x$ over and over. Each trigonometric identity links two of these functions, so it turns into a link between their coefficients. Once we compare coefficients of x^n, all three series are fixed by the single sequence b_n. The Pythagorean identity $\cot^2 x + 1 = \operatorname{cosec}^2 x$ then gives equations that the b_n must satisfy, and we can solve them one at a time.

Solution

Part (i)

A power series that converges to a function near 0 is that function's Maclaurin series, so $a_n = \tan^{(n)}(0)/n!$. Now tan is an odd function: $\tan(-x) = -\tan x$. Differentiating an odd function gives an even function, and differentiating an even function gives an odd one. So every even-order derivative of tan is odd, and an odd function vanishes at 0. Hence $\tan^{(n)}(0) = 0$ for even n, and so $a_n = 0$ for even n. Equivalently, the series contains only odd powers of x. The same argument applies to $\cot x - \frac{1}{x} = \sum b_n x^n$, which is also odd, so $b_n = 0$ for even n. We will also get this from the relation below.

For the identity, put everything over $\sin x \cos x$:

$$\cot x - \tan x = \frac{\cos^2 x - \sin^2 x}{\sin x \cos x} = \frac{\cos 2x}{\frac{1}{2}\sin 2x} = 2\cot 2x.$$

Now take $0 < x < \frac{1}{4}\pi$, so that $2x$ also lies in $(0, \frac{1}{2}\pi)$ and we may use the series for $\cot 2x$, with $2x$ in place of x:

$$2\cot 2x = 2\left(\frac{1}{2x} + \sum_{n=0}^{\infty} b_n 2^n x^n\right) = \frac{1}{x} + \sum_{n=0}^{\infty} 2^{n+1} b_n x^n.$$

The left-hand side of the identity is

$$\cot x - \tan x = \frac{1}{x} + \sum_{n=0}^{\infty} (b_n - a_n) x^n.$$

The $\frac{1}{x}$ terms cancel. We are left with two power series that are equal for every x in $(0, \frac{1}{4}\pi)$.

Ask yourself Why are we allowed to compare coefficients of two infinite series?

If two power series have the same sum throughout an interval, their coefficients agree. Subtract them: the difference is a power series that is zero on the whole interval. Then all its derivatives are zero there, and letting $x \to 0^+$ shows that every coefficient is zero. So

$$b_n - a_n = 2^{n+1} b_n, \qquad \text{that is,} \qquad a_n = (1 - 2^{n+1})\, b_n,$$

as required. Since $1 - 2^{n+1} \neq 0$, we have $b_n = a_n/(1 - 2^{n+1})$, so $b_n = 0$ whenever $a_n = 0$, which is for even n.

Part (ii)

In the same way,

$$\cot x + \tan x = \frac{\cos^2 x + \sin^2 x}{\sin x \cos x} = \frac{1}{\frac{1}{2}\sin 2x} = 2 \operatorname{cosec} 2x.$$

For $0 < x < \frac{1}{4}\pi$, the series for cosec with $2x$ in place of x gives

$$2 \operatorname{cosec} 2x = \frac{1}{x} + \sum_{n=0}^{\infty} 2^{n+1} c_n x^n, \qquad \cot x + \tan x = \frac{1}{x} + \sum_{n=0}^{\infty} (b_n + a_n) x^n.$$

Comparing coefficients gives $b_n + a_n = 2^{n+1} c_n$. Substituting $a_n = (1 - 2^{n+1}) b_n$ from part (i),

$$2^{n+1} c_n = (2 - 2^{n+1})\, b_n \quad \Longrightarrow \quad c_n = \frac{2 - 2^{n+1}}{2^{n+1}}\, b_n = (2^{-n} - 1)\, b_n,$$

which is the required result. In particular $c_n = 0$ for even n as well.

Part (iii)

Multiplying the series for $\cot x$ and $\operatorname{cosec} x$ by x,

$$1 + x \sum_{n=0}^{\infty} b_n x^n = x \cot x, \qquad 1 + x \sum_{n=0}^{\infty} c_n x^n = x \operatorname{cosec} x.$$

So the claimed equation says $x^2 \cot^2 x + x^2 = x^2 \operatorname{cosec}^2 x$. This is x^2 times the identity $\cot^2 x + 1 = \operatorname{cosec}^2 x$, which comes from dividing $\cos^2 x + \sin^2 x = 1$ by $\sin^2 x$. That proves it.

Now we expand. The even-indexed b_n and c_n are zero, so

$$x\cot x = 1 + b_1x^2 + b_3x^4 + \cdots, \qquad x\operatorname{cosec} x = 1 + c_1x^2 + c_3x^4 + \cdots,$$

and the equation becomes

$$(1 + b_1x^2 + b_3x^4 + \cdots)^2 + x^2 = (1 + c_1x^2 + c_3x^4 + \cdots)^2.$$

Squaring,

$$1 + (2b_1 + 1)x^2 + (2b_3 + b_1^2)x^4 + \cdots = 1 + 2c_1x^2 + (2c_3 + c_1^2)x^4 + \cdots.$$

From part (ii), $c_1 = (\frac{1}{2} - 1)b_1 = -\frac{1}{2}b_1$ and $c_3 = (\frac{1}{8} - 1)b_3 = -\frac{7}{8}b_3$.

Coefficient of x^2:

$$2b_1 + 1 = 2c_1 = -b_1 \quad \Longrightarrow \quad b_1 = -\tfrac{1}{3}.$$

Then part (i) gives $a_1 = (1 - 2^2)b_1 = (-3)(-\frac{1}{3}) = 1$, as required.

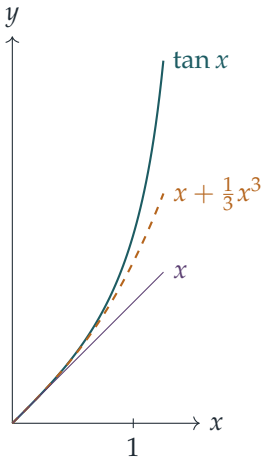
Coefficient of x^4: with $b_1 = -\frac{1}{3}$ and $c_1 = \frac{1}{6}$,

$$2b_3 + \tfrac{1}{9} = -\tfrac{7}{4}b_3 + \tfrac{1}{36} \quad \Longrightarrow \quad \tfrac{15}{4}\,b_3 = -\tfrac{3}{36} = -\tfrac{1}{12} \quad \Longrightarrow \quad b_3 = -\tfrac{1}{45}.$$

Hence

$$a_3 = (1 - 2^4)\,b_3 = (-15)\left(-\tfrac{1}{45}\right) = \tfrac{1}{3}.$$

So $\tan x = x + \frac{1}{3}x^3 + \cdots$.



Checks

- At $x = 0.3$, $\tan 0.3 \approx 0.309336$, and $0.3 + \frac{1}{3}(0.3)^3 = 0.309$. The difference is about 3×10^{-4}, which is the size of the next term, $\frac{2}{15}x^5$.
- The value $b_1 = -\frac{1}{3}$ matches the familiar $\cot x \approx \frac{1}{x} - \frac{x}{3}$. Also $c_1 = \frac{1}{6}$ matches $\operatorname{cosec} x \approx \frac{1}{x} + \frac{x}{6}$, which you can get from $\sin x \approx x - \frac{x^3}{6}$.

- We checked all three relations exactly (in rational arithmetic) for every n up to 48 by dividing the series for sin and cos. We also checked that the partial sums agree with tan, cot and cosec at random points in $(0, 1)$.

Common slips

- Comparing coefficients while forgetting the $\frac{1}{x}$ terms. They cancel, but only because of the factor 2 in $2\cot 2x$ and $2\operatorname{cosec} 2x$.
- Forgetting that $b_0 = b_2 = 0$ when expanding the squares in part (iii). Keeping b_0 adds spurious odd powers of x, and you would have to show separately that their coefficients vanish.
- Getting $c_3 = -\frac{7}{8}b_3$ with the wrong sign or power of 2. Note that $2^{-3} - 1 = -\frac{7}{8}$.

Beyond STEP

Work through this method to higher order and the coefficients turn out to be the Bernoulli numbers B_{2k}: $x\cot x = \sum_k (-1)^k 2^{2k} B_{2k}\, x^{2k}/(2k)!$. The factor $1 - 2^{n+1}$ from part (i) then gives the general formula for the coefficients of $\tan x$. Why does the series for $\tan x$ stop converging at $|x| = \frac{1}{2}\pi$? Complex analysis answers this: a power series converges out to the nearest point where the function blows up, which for tan is the pole at $\pm\frac{1}{2}\pi$. Needham's *Visual Complex Analysis* explains this beautifully, and Abbott's *Understanding Analysis* makes the term-by-term comparison of power series rigorous.

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